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BATCH- CH2

PRACTICAL NO.3

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`
- The shape of the array using `shape`
- The total number of elements in the array using `size` Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next lines contain space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array

CODE:

```
import numpy as np
```

```
r, c = map(int, input().split())
```

```
elements = [] for i
```

```
in range(r):
```

```
    row = list(map(int, input().split()))
```

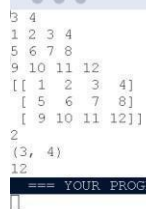
```
elements.extend(row)
```

```
arr = np.array(elements).reshape(r, c)
```

```
print(arr)    print(arr.ndim)    print(arr.shape)
```

```
print(arr.size)
```

OUTPUT:



```
3 4
1 2 3 4
5 6 7 8
9 10 11 12
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]
2
(3, 4)
12
=== YOUR PROGRAM HAS ENDED ===
```

Terminal Test cases

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

1. **Addition** (`matrix_a + matrix_b`)
2. **Subtraction** (`matrix_a - matrix_b`)
3. **Element-wise Multiplication** (`matrix_a * matrix_b`)
4. **Matrix Multiplication** (`matrix_a · matrix_b`)
5. **Transpose of Matrix A**

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

1. The result of Addition.
2. The result of Subtraction.
3. The result of Element-wise Multiplication.
4. The result of Matrix Multiplication.

CODE:

```
import numpy as np
```

```
# Input matrices
print("Enter Matrix A:")
matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
```

```
print("Enter Matrix B:")
```

```
matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Addition
```

```
addition_result = matrix_a + matrix_b
```

```
print("Addition (A + B):")
print(addition_result)
```

```
# Subtraction
subtraction_result = matrix_a -
```

```
matrix_b
print("Subtraction (A - B):")
print(subtraction_result)
#
```

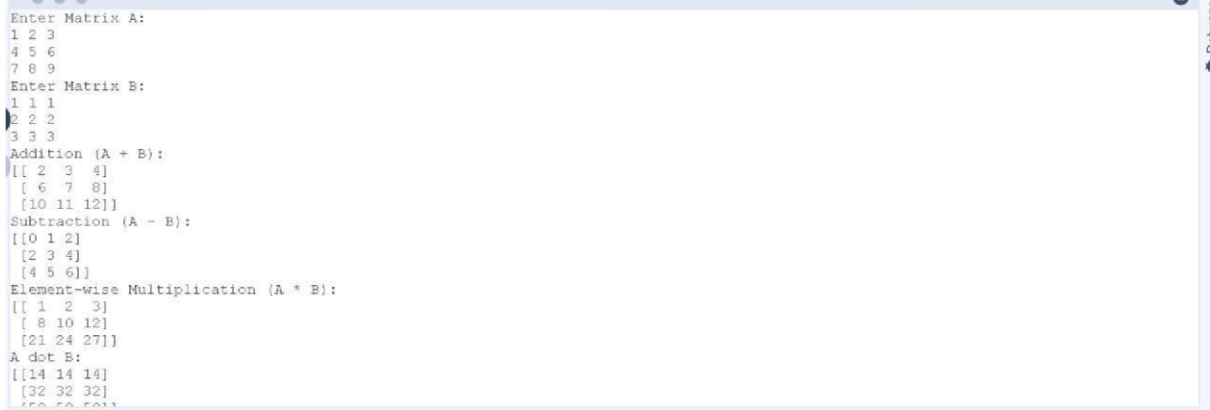
```

Multiplication (element-wise) elementwise_multiplication_result
= matrix_a * matrix_b
print("Element-wise Multiplication (A * B):")
print(elementwise_multiplication_result) # Matrix multiplication
(dot product) matrix_multiplication_result =
np.dot(matrix_a, matrix_b) print("A dot B:")
print(matrix_multiplication_result)

# Transpose transpose_a =
matrix_a.T print("Transpose
of A:") print(transpose_a)

```

OUTPUT:



```

Enter Matrix A:
1 2 3
4 5 6
7 8 9
Enter Matrix B:
1 1 1
2 2 2
3 3 3
Addition (A + B):
[[ 2  3  4]
 [ 6  7  8]
 [10 11 12]]
Subtraction (A - B):
[[ 0  1  2]
 [ 2  3  4]
 [ 4  5  6]]
Element-wise Multiplication (A * B):
[[ 1  2  3]
 [ 8 10 12]
 [21 24 27]]
A dot B:
[[14 14 14]
 [32 32 32]
 [50 50 50]]

```

```
A dot B:  
[[14 14 14]  
 [32 32 32]  
 [50 50 50]]  
Transpose of A:  
[[1 4 7]  
 [2 5 8]  
 [3 6 9]]  
=== YOUR PROGRAM HAS ENDED ===
```

You are given two arrays arr1 and arr2. You need to perform horizontal and vertical stacking operations on them using NumPy.

- **Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- **Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (sidebyside stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

CODE:

```
import numpy as np
```

```
# Input matrices print("Enter Array1:") arr1 = np.array([list(map(int,
input().split())) for i in range(3)])
```

```
print("Enter Array2:")
```

```
arr2 = np.array([list(map(int, input().split())) for i in range(3)])
```

```
# Perform horizontal stacking (hstack) h_stack
= np.hstack((arr1, arr2))
```

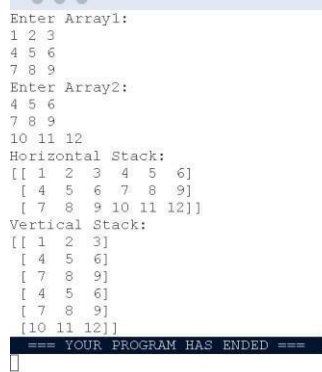
```
# Perform vertical stacking (vstack) v_stack
= np.vstack((arr1, arr2))
```

```
#      Output      results
```

```
print("Horizontal Stack:") print(h_stack)
```

```
print("Vertical Stack:") print(v_stack)
```

OUTPUT:



```
Enter Array1:
1 2 3
4 5 6
7 8 9
Enter Array2:
4 5 6
7 8 9
10 11 12
Horizontal Stack:
[[ 1  2  3  4  5  6]
 [ 4  5  6  7  8  9]
 [ 7  8  9 10 11 12]]
Vertical Stack:
[[ 1  2  3]
 [ 4  5  6]
 [ 7  8  9]
 [ 4  5  6]
 [ 7  8  9]
 [10 11 12]]
=== YOUR PROGRAM HAS ENDED ===
```

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

CODE:

```
import numpy as np
```

```
# Take user input for the start, stop, and step of the sequence
```

```
start = int(input()) stop = int(input()) step = int(input())
```

```
# Generate the sequence using np.arange()
```

```
sequence = np.arange(start, stop, step)
```

```
# Print the generated sequence
```

```
print(sequence)
```

OUTPUT:

A screenshot of a code editor window. The editor shows a Python script with the following content:

```
3  
10  
2  
[3 5 7 9]  
=== YOUR PROGRAM HAS ENDED ===
```

 The text "=== YOUR PROGRAM HAS ENDED ===" is highlighted in a dark blue background. At the bottom of the editor, there are two tabs: "Terminal" and "Test cases".

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_i OR B_i).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

CODE:

```
import numpy as np

def array_operations(A, B):

    # Convert A and B to NumPy arrays
    A = np.array(A)
    B = np.array(B)

    # Arithmetic Operations
    sum_result = A + B
    diff_result = A - B
    prod_result = A * B

    # Statistical Operations
    mean_A = np.mean(A)
    median_A = np.median(A)
    std_dev_A = np.std(A)

    # Bitwise Operations
    and_result = A & B
```

```
or_result = A | B xor_result = A
```

```
^ B
```

```
# Output results with one space between each element print("Elementwise  
Sum:", ' '.join(map(str, sum_result))) print("Element-wise Difference:", '  
' '.join(map(str, diff_result))) print("Element-wise Product:", ' '.join(map(str,  
prod_result)))
```

```
print(f"Mean of A: {mean_A}")
```

```
print(f"Median of A: {median_A}") print(f"Standard  
Deviation of A: {std_dev_A}") print("Bitwise AND:", '  
' '.join(map(str, and_result))) print("Bitwise OR:", '  
' '.join(map(str, or_result))) print("Bitwise XOR:", '  
' '.join(map(str, xor_result)))
```

```
A = list(map(int, input().split())) # Elements of array
```

```
A B = list(map(int, input().split())) # Elements of
```

```
array B array_operations(A, B)
```

OUTPUT:

```
1 2 3 4  
5 6 7 8  
Element-wise Sum: 6 8 10 12  
Element-wise Difference: -4 -4 -4 -4  
Element-wise Product: 5 12 21 32  
Mean of A: 2.5  
Median of A: 2.5  
Standard Deviation of A: 1.118033988749895  
Bitwise AND: 1 2 3 0  
Bitwise OR: 5 6 7 12  
Bitwise XOR: 4 4 4 12  
=== YOUR PROGRAM HAS ENDED ===
```

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>

View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>

Copy array: <copy_array>

CODE:

```
import numpy as np
```

```
inputlist = list(map(int,input().split(" ")))
```

```
# Original array
```

```
original_array = np.array(inputlist)
```

```
# Create a view
```

```
view_array = original_array.view()
```

```
# Create a copy
```

```
copy_array = original_array.copy()
```

```
# Modify the view view_array[0]
```

```
= 99 print("Original array after modifying view:", original_array) print("View  
array:", view_array)
```

```
# Modify the copy copy_array[1]
```

```
= 88 print("Original array after modifying copy:", original_array) print("Copy  
array:", copy_array)
```

OUTPUT:



```
10 20 30 40 50 60 70 80
Original array after modifying view: [99 20 30 40 50 60 70 80]
View array: [99 20 30 40 50 60 70 80]
Original array after modifying copy: [99 20 30 40 50 60 70 80]
Copy array: [10 88 30 40 50 60 70 80]
=== YOUR PROGRAM HAS ENDED ===
```

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.

- `broadcast_value`: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- 1. Searching:** Find the indices where `search_value` appears in `array1` and print these indices.
- 2. Counting:** Count how many times `count_value` appears in `array1` and print the count.
- 3. Broadcasting:** Add `broadcast_value` to each element of `array1` using broadcasting, and print the resulting array.
- 4. Sorting:** Sort `array1` in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing `array1`.
2. An integer `search_value` represents the value to search for in the array.
3. An integer `count_value` represents the value to count in the array.
4. An integer `broadcast_value` represents the value to add to each element of the array.

Output Format:

1. The indices where `search_value` occurs in `array1`.
2. The count of occurrences of `count_value` in `array1`.
3. The array after adding the `broadcast_value` to each element.
4. The sorted array.

CODE:

```
import numpy as np
```

```
# Input array from the user array1 = np.array(list(map(int,
input().split()))))
```

Searching

```
search_value = int(input("Value to search: ")) count_value  
= int(input("Value to count: ")) broadcast_value =  
int(input("Value to add: "))
```

Find indices where value matches in array1 indices

```
= np.where(array1 == search_value)[0] print(indices)
```

Count occurrences in array1 count =

```
np.sum(array1 == count_value) print(count)
```

Broadcasting addition broadcasted = array1

```
+ broadcast_value
```

```
print(broadcasted) # Sort the first array
```

```
sorted_array = np.sort(array1) print(sorted_array)
```

OUTPUT:



```
1 1 1 2 2 2  
Value to search: 1  
Value to count: 2  
Value to add: 2  
[0 1 2]  
3  
[3 3 3 4 4 4]  
[1 1 1 2 2 2]  
=== YOUR PROGRAM HAS ENDED ===  
Terminal Test cases
```

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.

- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who got less than 40 marks in Subject 1:** Count the number of students who scored less than 40 marks in Subject 1.
- **Find the count of students who got more than 90 marks in Subject 2:** Count the number of students who scored more than 90 marks in Subject 2.
- **Find the count of students who scored ≥ 90 in each subject:** Count the number of students who scored 90 or more marks in each subject.
- **Find the count of subjects in which each student scored ≥ 90 :** Determine how many subjects each student scored 90 or more marks in.
- **Print Subject 1 marks in ascending order:** Sort and print the marks of students in Subject 1 in ascending order.
- **Print students who scored between 50 and 90 in Subject 1:** Display students who scored marks between 50 and 90 in Subject 1.
- **Find index positions of students who scored 79 in Subject 1:** Identify the index positions of students who scored exactly 79 marks in Subject 1.

CODE:

```
import numpy as np
```



```

a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)

# 1
print("All student Details:\n", a)

# 2
print("Total Students:", a.shape[0])

# 3
print("All Student Roll Nos", a[:, 0])

# 4
print("Subject 1 Marks", a[:, 1])

# 5
print("Min marks in Subject 2", np.min(a[:, 2]))

# 6
print("Max marks in Subject 3", np.max(a[:, 3]))

# 7
print("All subject marks:", a[:, 1:])

# 8
total = np.sum(a[:, 1:], axis=1) print("Total Marks",
total) # 9 (no label in sample output)

print(np.round(np.mean(a[:, 1:], axis=1), 1)) # 10

print("Average Marks of each subject", np.round(np.mean(a[:, 1:],
axis=0), 1)) # 11 print("Average Marks of S1 and S2",
np.round(np.mean(a[:, [1,
2]], axis=0), 1)) # 12 print("Average Marks of S1 and S3",
np.round(np.mean(a[:, [1, 3]],
axis=0), 1)) #

13 print("Roll no who got maximum marks in Subject 3",
a[np.argmax(a[:, 3]), 0]) # 14 print("Roll no who got minimum marks

```

```

in Subject 2", a[np.argmin(a[:, 2]), 0]) # 15 print("Roll no who got 24
marks in
Subject 2", a[a[:, 2] == 24, 0].reshape(-1,1))
# 16
print("Count of students who got marks in Subject 1 < 40", np.sum(a[:, 1]
< 40)) # 17 print("Count of students who got marks
in Subject 2 > 90:", np.sum(a[:, 2] > 90)) # 18
print("Count of students in each subject who got marks >= 90:",
np.sum(a[:, 1:] >= 90, axis=0))
# 19
print("Roll no:", a[:, 0])
print("Count of subjects in which student got marks >= 90:",
np.sum(a[:, 1:] >= 90, axis=1))
# 20
print(np.sort(a[:, 1]))
# 21 (print full rows like sample) print(a[(a[:,
1] >= 50) & (a[:, 1] <= 90)]) print(a) # 22
(exact output format)
print(np.where(a[:, 1] == 79))

```

OUTPUT:

```

All student Details:
[[301. 67. 77. 88.]
 [302. 78. 88. 77.]
 [303. 45. 56. 89.]
 [304. 88. 98. 45.]
 [305. 78. 88. 99.]
 [306. 68. 78. 36.]
 [307. 79. 12. 45.]
 [308. 46. 45. 77.]
 [309. 89. 32. 88.]
 [310. 79. 24. 33.]]
Total Students: 10
All Student Roll Nos [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Subject 1 Marks [67. 78. 45. 88. 78. 68. 79. 46. 89. 79.]
Min marks in Subject 2 12.0
Max marks in Subject 3 99.0
All subject marks: [[67. 77. 88.]
 [78. 88. 77.]
 [45. 56. 89.]
 [88. 98. 45.]
 [78. 88. 99.]
 [68. 78. 36.]
 [79. 12. 45.]
 [46. 45. 77.]
 [89. 32. 88.]
 [79. 24. 33.]]
Total Marks [232. 243. 190. 231. 265. 182. 136. 168. 209. 136.]
[77.3 81. 63.3 77. 88.3 60.7 45.3 56. 69.7 45.3]
Average Marks of each subject [71.7 59.8 67.7]
Average Marks of S1 and S2 [71.7 59.8]
Average Marks of S1 and S3 [71.7 67.7]
Roll no who got maximum marks in Subject 3 305.0
Roll no who got minimum marks in Subject 2 307.0
Roll no who got 24 marks in Subject 2 [[310.]]
Count of students who got marks in Subject 1 < 40 0
Count of students who got marks in Subject 2 > 90: 1
Count of students in each subject who got marks >= 90: [0 1 1]
Roll no: [301. 302. 303. 304. 305. 306. 307. 308. 309. 310.]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
Count of subjects in which student got marks >= 90: [0 0 0 1 1 0 0 0 0 0]
[45. 46. 67. 68. 78. 78. 79. 79. 88. 89.]
[[301. 67. 77. 88.]
 [302. 78. 88. 77.]
 [304. 88. 98. 45.]
 [305. 78. 88. 99.]
 [306. 68. 78. 36.]
 [307. 79. 12. 45.]
 [309. 89. 32. 88.]
 [310. 79. 24. 33.]]
[[301. 67. 77. 88.]
 [302. 78. 88. 77.]
 [303. 45. 56. 89.]
 [304. 88. 98. 45.]
 [305. 78. 88. 99.]
 [306. 68. 78. 36.]
 [307. 79. 12. 45.]
 [308. 46. 45. 77.]
 [309. 89. 32. 88.]
 [310. 79. 24. 33.]]
(array([6, 9], dtype=int32),)

```

=== YOUR PROGRAM HAS ENDED ===



Terminal



Test cases